



Sharif University of Technology

Department of Computer Engineering

Master's Seminar – Artificial Intelligence and Robotics

Predicting Catastrophic Overfitting in Fast Adversarial Training

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Section 1

Introduction

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Adversarial Examples

- Deep neural networks achieve remarkable performance across vision, NLP, and autonomous systems.
- However, they are highly vulnerable to **adversarial attacks**: small, often imperceptible perturbations that flip predictions with high confidence [1].
- This vulnerability raises serious concerns for safety-critical applications.

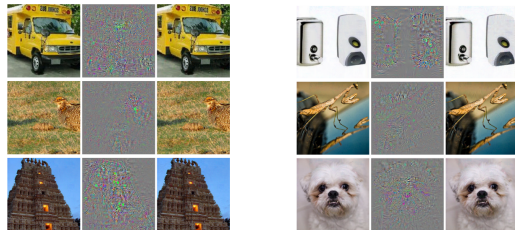


Figure 1: Adversarial example for AlexNet [2, 1]

- **Adversarial Training (AT)**: the leading defense; trains on adversarially perturbed samples instead of clean data [3].
- Multi-step AT is effective but **computationally expensive** – limits scalability.
- **Fast Adversarial Training (FAT)** uses single-step attacks to reduce cost [4].
- FAT suffers from **Catastrophic Overfitting (CO)**: robust accuracy against multi-step attacks suddenly collapses to near zero after a few epochs, while single-step accuracy keeps rising.

Why Does This Matter?

- Robustness is critical in security-sensitive applications: autonomous driving, medical diagnosis, and finance.
- Multi-step AT is reliable but too costly for large-scale models/datasets.
- FAT is efficient, but **unreliable** until CO is solved.
- Most existing CO-prevention methods add expensive hyperparameter tuning or extra computation.

Goal of this seminar

Review key concepts and prior work on catastrophic overfitting, and introduce an early idea for **low-cost prediction** of CO and **dynamic step-size** adjustment.

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Fast Gradient Sign Method (FGSM) [5]:

$$\delta = \epsilon \cdot \text{sign}(\nabla_x \mathcal{L}(f_\theta(x), y))$$

Projected Gradient Descent (PGD) [3]:

$$x^{t+1} = \Pi_{\|x' - x\|_\infty \leq \epsilon} \left(x^t + \alpha \cdot \text{sign}(\nabla_x \mathcal{L}(f_\theta(x^t), y)) \right)$$

- ϵ : perturbation radius (e.g., 8/255) α : step size (e.g., 2/255)
- PGD-10 / PGD-20: iterated for 10 / 20 steps → **robust accuracy**

Min-max formulation [3]:

$$\min_{\theta} \mathbb{E}_{(x,y) \sim \mathcal{D}} \left[\max_{\|\delta\|_{\infty} \leq \epsilon} \mathcal{L}(f_{\theta}(x + \delta), y) \right]$$

- Inner max: find worst-case perturbation.
- Outer min: update model to resist it.
- Standard AT solves the inner problem with multi-step PGD $\rightarrow$ up to $10\times$ training cost.
- **Fast AT** approximates the inner max with a *single* FGSM step [4].

Catastrophic Overfitting (CO)

- After a few epochs of single-step FAT, **PGD robust accuracy collapses** to near zero.
- **FGSM accuracy** keeps increasing – sometimes even above clean accuracy.
- The model overfits to the specific attack used in training rather than gaining true robustness.

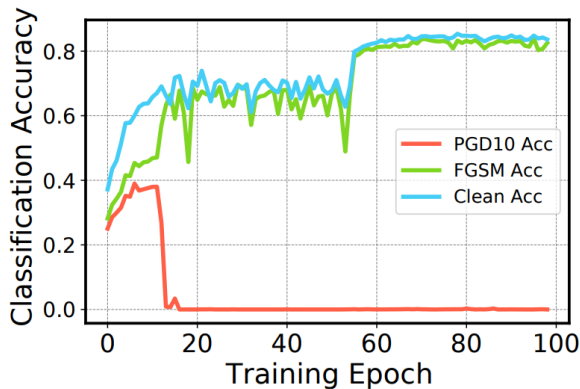


Figure 2: CO in FAT: clean / FGSM / PGD-10 accuracy over training [6]

- Smoother (lower-curvature) loss $\Rightarrow$ single-step attacks approximate the true maximum better.
- **Evaluation:** FGSM accuracy alone is misleading; reliable evaluation uses strong multi-step attacks such as **AutoAttack** [7].
- Key trade-off: **robustness vs. clean accuracy** [8].

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Prior Work

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Randomization-Based Methods

- **FGSM-RS** [4]: random start before FGSM perturbation – largely prevents CO but not always at larger ϵ .
- **N-FGSM** [9]: larger-magnitude random noise (radius $k > \epsilon$) with *no clipping* $\rightarrow$ more diverse perturbations, better robustness.

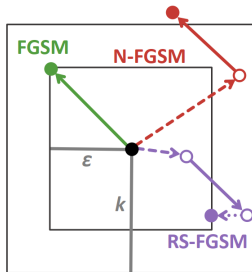


Figure 3: FGSM vs. RS-FGSM vs. N-FGSM [5, 4, 9]

Regularization-Based Methods

- **GradAlign** [10]: maximizes gradient alignment between clean and perturbed samples.
- **NuAT** [11]: nuclear-norm regularization to smooth the loss.
- **AAER** [12]: detects and regularizes *abnormal adversarial examples*.
- **ELLE** [13]: encourages local linearity of the loss surface efficiently.

$$\mathcal{L}_{\text{GradAlign}}(\theta) = \underbrace{\mathcal{L}(f_{\theta}(x + \delta_{\text{FGSM}}), y)}_{\text{standard FAT loss}} + \lambda \underbrace{\mathbb{E}_{\eta \sim \mathcal{U}([- \epsilon, \epsilon]^d)} \left[1 - \cos(\nabla_x \mathcal{L}(x, y; \theta), \nabla_x \mathcal{L}(x + \eta, y; \theta)) \right]}_{\text{gradient-alignment penalty}}$$

λ : regularization strength. Penalizing $1 - \cos(\cdot, \cdot)$ pushes the alignment toward 1 $\Rightarrow$ a more locally-linear loss, directly counteracting the curvature growth behind CO.

Adaptive step size:

- Reinforcement-learning-based step size [14]
- **ATAS** – step size adapted via gradient norm [15]
- Per-sample step size from noise/gradient similarity [16]

Other approaches:

- **ZeroGrad** [17]: drops updates with negligible gradient magnitude.
- **Free AT** [18]: reuses attack gradients for weight updates.
- **Curvature regularization** [19]: directly controls loss curvature.

No metric has been designed *purely* for low-cost CO prediction, but a few serve as indirect early signals:

Gradient Alignment [10]:

$$\mathbb{E}_{(x,y) \sim \mathcal{D}, \eta \sim \mathcal{U}([- \epsilon, \epsilon]^d)} [\cos(\nabla_x \mathcal{L}(x, y; \theta), \nabla_x \mathcal{L}(x + \eta, y; \theta))]$$

- Near 1 $\Rightarrow$ locally linear loss; sharp drop $\Rightarrow$ CO warning. **Costs an extra backward pass.**

Abnormal Adversarial Examples (AAE) [12]: samples where adding the FGSM perturbation *decreases* the loss – also costly to track (loss at two points).

Limitations of Prior Work

- ❶ **Dataset/architecture dependence** – tuned mainly for CIFAR-10-scale settings.
- ❷ **Hyperparameter tuning** required per dataset/architecture.
- ❸ **Extra computational cost** (e.g., GradAlign, ELLE).
- ❹ **No efficient prediction metric** exists that anticipates CO *before* it happens at negligible cost.

Gap

No prior method simultaneously solves all four issues – this motivates the approach explored next.

Comparison of Prior Methods (1/2)

Table 1: Comparison of prior methods for preventing/predicting catastrophic overfitting

Method	Strategy	Advantage	Limitation
FGSM-RS [4]	Randomization	Simple, near-zero overhead	Still prone to CO at larger ϵ
N-FGSM [9]	Randomization	Better robustness & accuracy	Noise scale needs tuning
GradAlign [10]	Regularization	Clear theory; usable as CO predictor	Extra backward pass; high cost
NuAT [11]	Regularization	Improves training stability	Extra compute & memory
AAER [12]	Regularization	Targets a root cause of CO	Multi-point loss eval; threshold-sensitive
ELLE [13]	Regularization	Strong robustness on hard datasets	Higher compute/memory than baselines

Comparison of Prior Methods (2/2)

Table 2: Comparison of prior methods (cont'd)

Method	Strategy	Advantage	Limitation
RL-based [14]	Adaptive step	Adapts to training dynamics	Complex; costly RL training
ATAS [15]	Adaptive step	Efficient; no extra learned component	Depends on gradient-norm estimate quality
Similarity-based [16]	Adaptive step	Per-sample granularity	Extra per-sample computation
ZeroGrad [17]	Other	Very simple, low-cost	Limited effectiveness on harder datasets
Free AT [18]	Other	Removes separate attack overhead	Lower accuracy/robustness than specialized methods
Curvature Reg. [19]	2nd-order	Strong theoretical link to robustness	Rarely used in fast/cheap AT setting

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Observation: Epsilon Overfitting

- Trained PreActResNet18 on CIFAR-10 with fixed $\epsilon = 8/255$ until CO occurred.
- Measured accuracy vs. attack strength ($-16/255$ to $+16/255$) for FGSM and PGD-10.
- FGSM accuracy is highly **erratic** across ϵ values; PGD-10 stays near zero throughout.

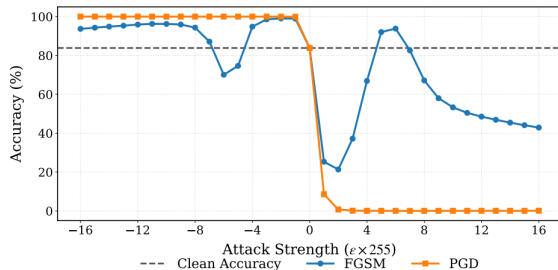


Figure 4: Accuracy vs. attack strength for a CO model

→ The model overfits to specific perturbation magnitudes (“**epsilon overfitting**”) rather than gaining true robustness – motivating diversity in perturbation size/direction.

Prediction Metric: Perturbation Alignment

$$\text{PertAlign} = \cos\left(\nabla_x \mathcal{L}(f_\theta(x'), y), \nabla_x \mathcal{L}(f_\theta(x' + \delta), y)\right)$$

where $x' = x + \eta$, $\eta \sim \mathcal{U}([- \epsilon, \epsilon]^d)$, and δ is a small step in the attack direction.

- Near 1 $\Rightarrow$ locally linear loss; drops toward 0 near CO onset.
- **Key advantage:** both gradients are already computed during standard single-step AT $\Rightarrow$ **virtually no extra cost.**

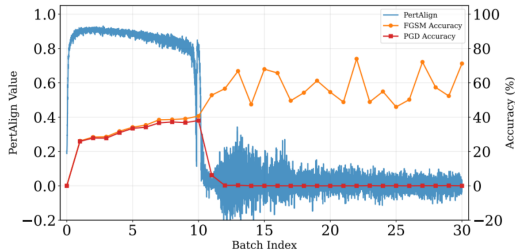


Figure 5: PertAlign drop precedes FGSM/PGD-10 divergence

Toward Dynamic Step-Size Adjustment

- View the inner maximization through a **second-order local approximation** of the loss.
- Use the same gradients already computed for PertAlign to estimate a locally optimal step size – **nearly free**.
- Combining: (1) diversified perturbation distribution, (2) low-cost CO prediction via PertAlign, and (3) dynamic step size → a method robust across datasets/architectures with a **fixed** hyperparameter set.
- Full method design and large-scale evaluation: **future thesis work**.

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Conclusion

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- Catastrophic overfitting is the main obstacle to efficient, reliable Fast Adversarial Training.
- Reviewed core concepts (attacks, AT, FAT, CO).
- Surveyed prior work: randomization, regularization, adaptive step size, and other strategies – each partially effective, none solving all four key limitations.
- Preliminary evidence presented:
 - **Epsilon overfitting** phenomenon in FGSM accuracy.
 - **PertAlign** metric providing an early, near-free warning signal for CO.
- Next step: full design and evaluation of a low-cost, dynamic-step-size FAT method.

Roadmap for Future Work

- Designing new ideas (e.g., dynamic step-size) and implementing them: ~ 3 months
- Evaluating and validating the proposed methods: ~ 2 months
- Thesis writing and project defense: ~ 3 months

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Thank You!

Questions & Discussion

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